



GF_VIP

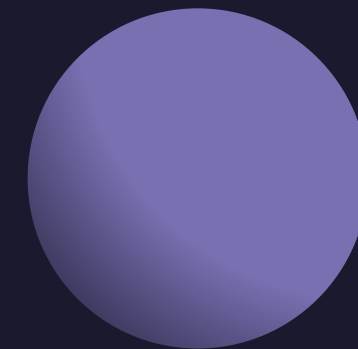
La tua privacy a portata di
blockchain

Francesco Petrillo
Gianfranco Vitiello



Introduzione

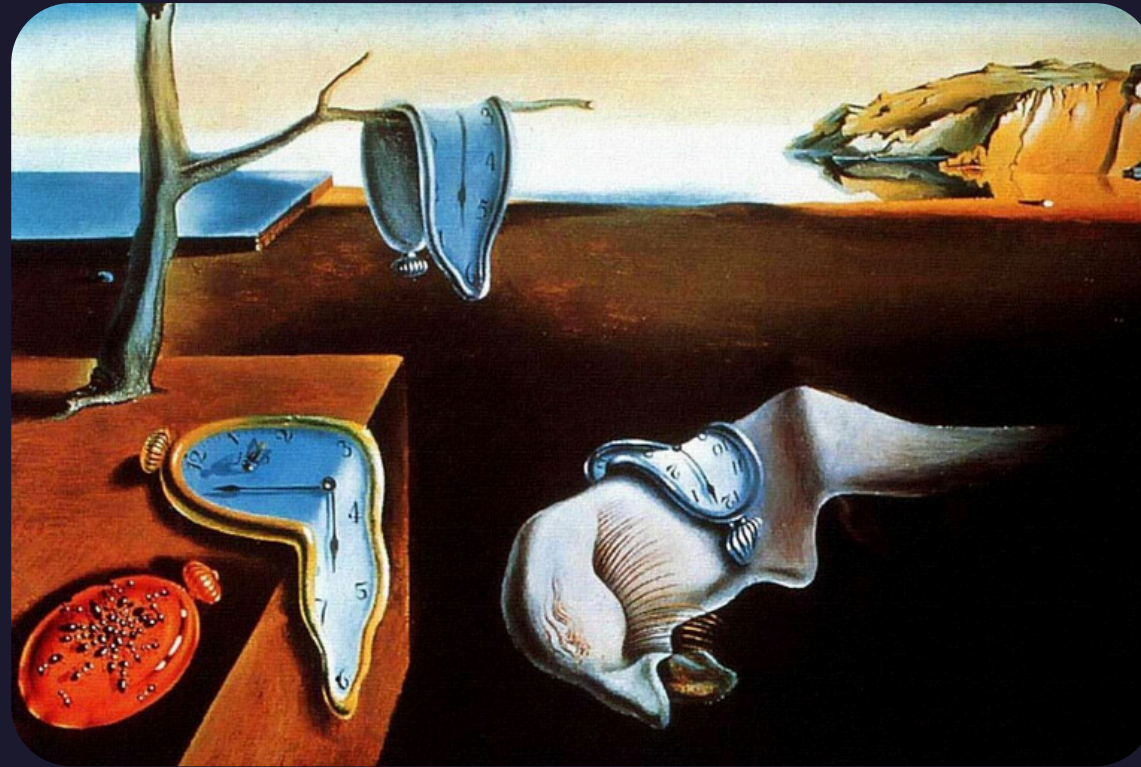
Il problema



Il problema

Verifica attuale delle competenze:

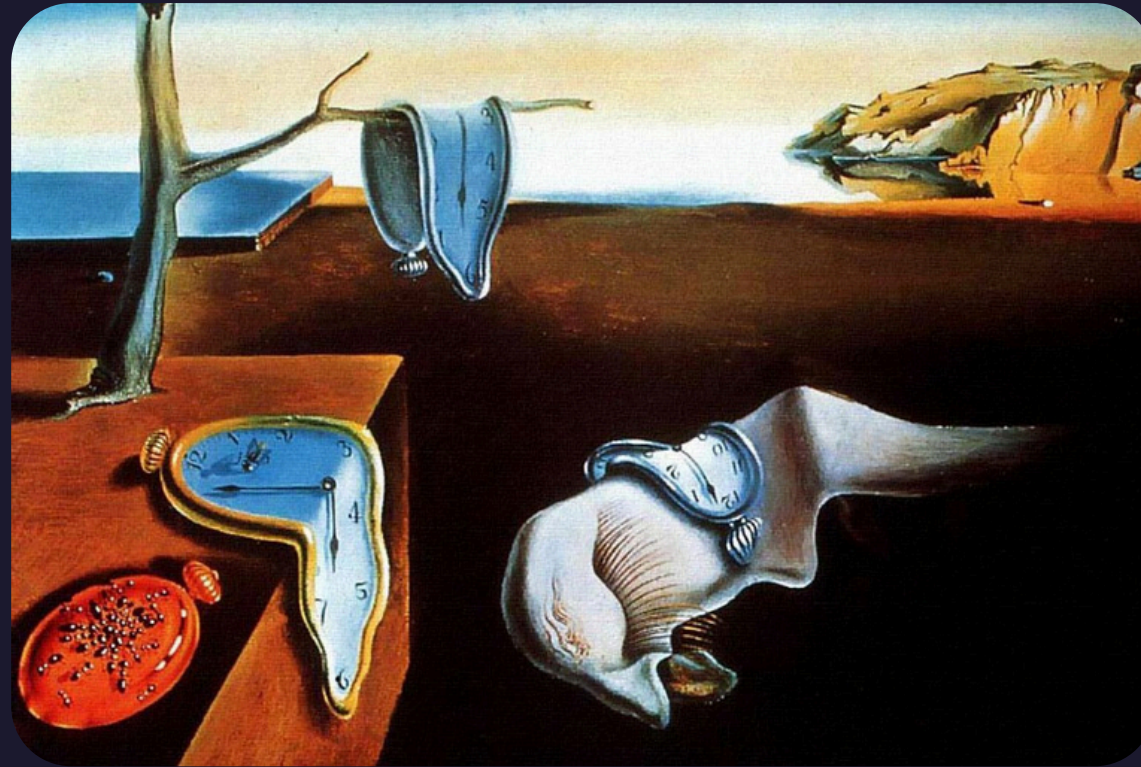
- Lungo



Il problema

Verifica attuale delle competenze:

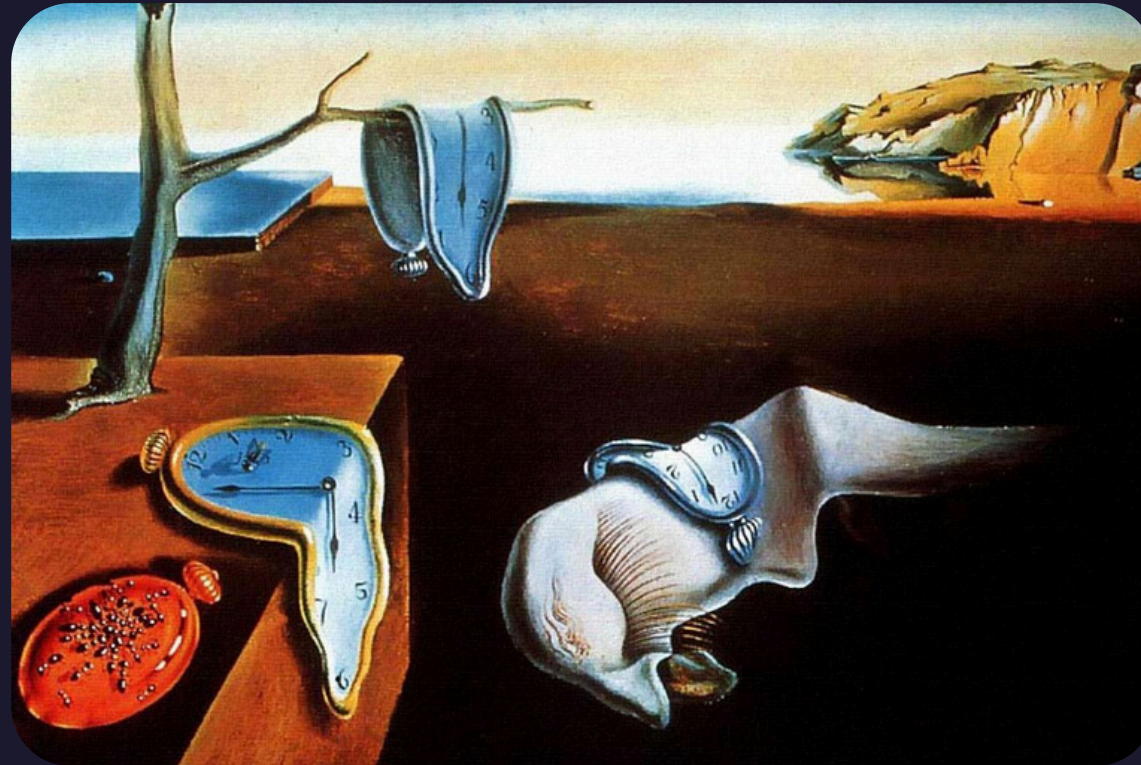
- Lungo
- Difficile



Il problema

Verifica attuale delle competenze:

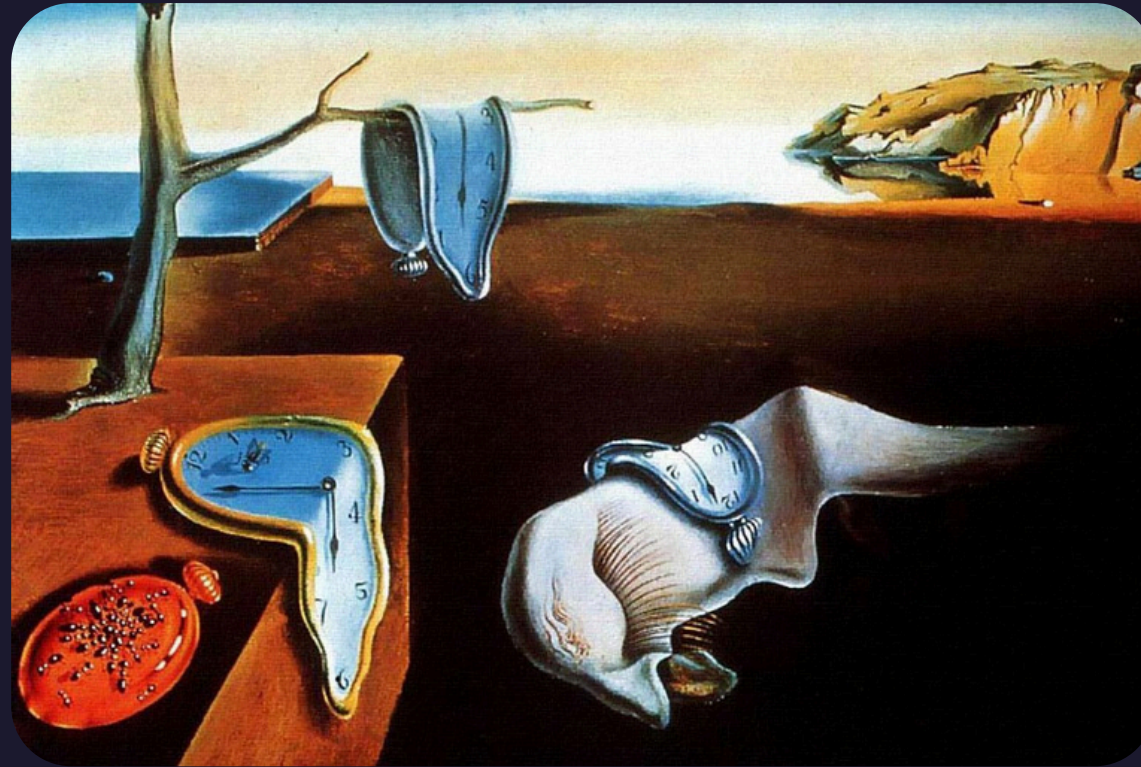
- Lungo
- Difficile
- Fiducia



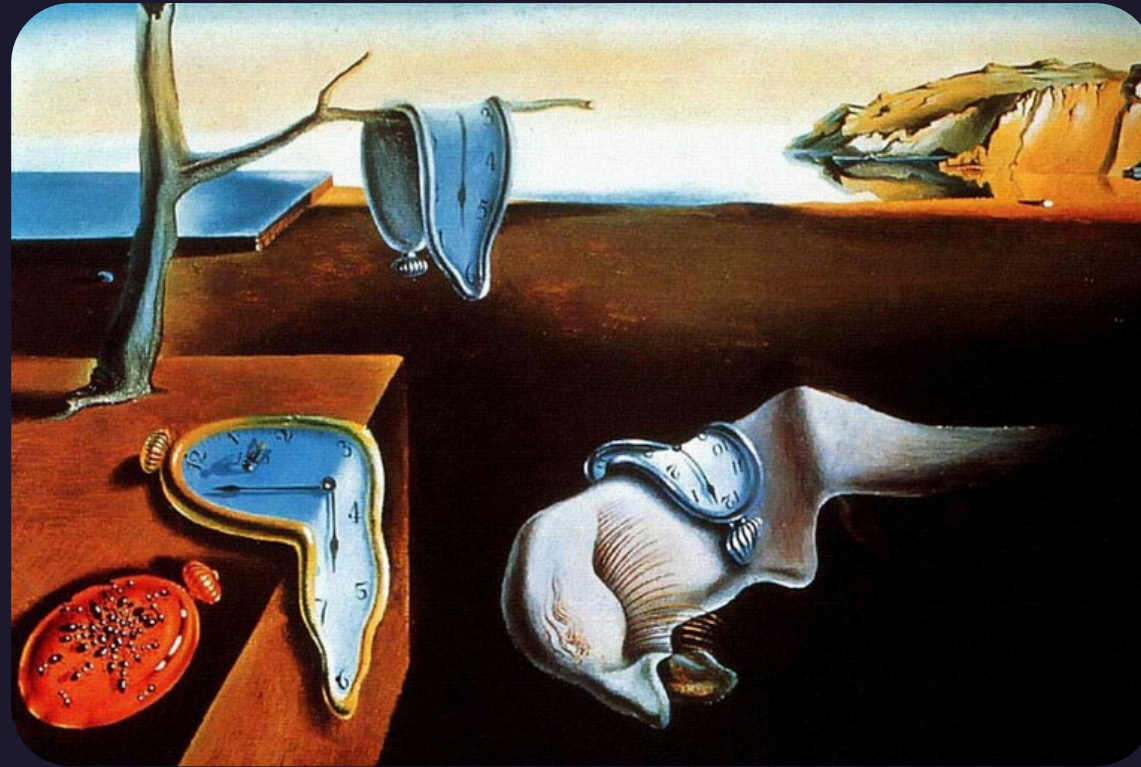
Il problema

Verifica attuale delle competenze:

- Lungo
- Difficile
- Fiducia
- Privacy

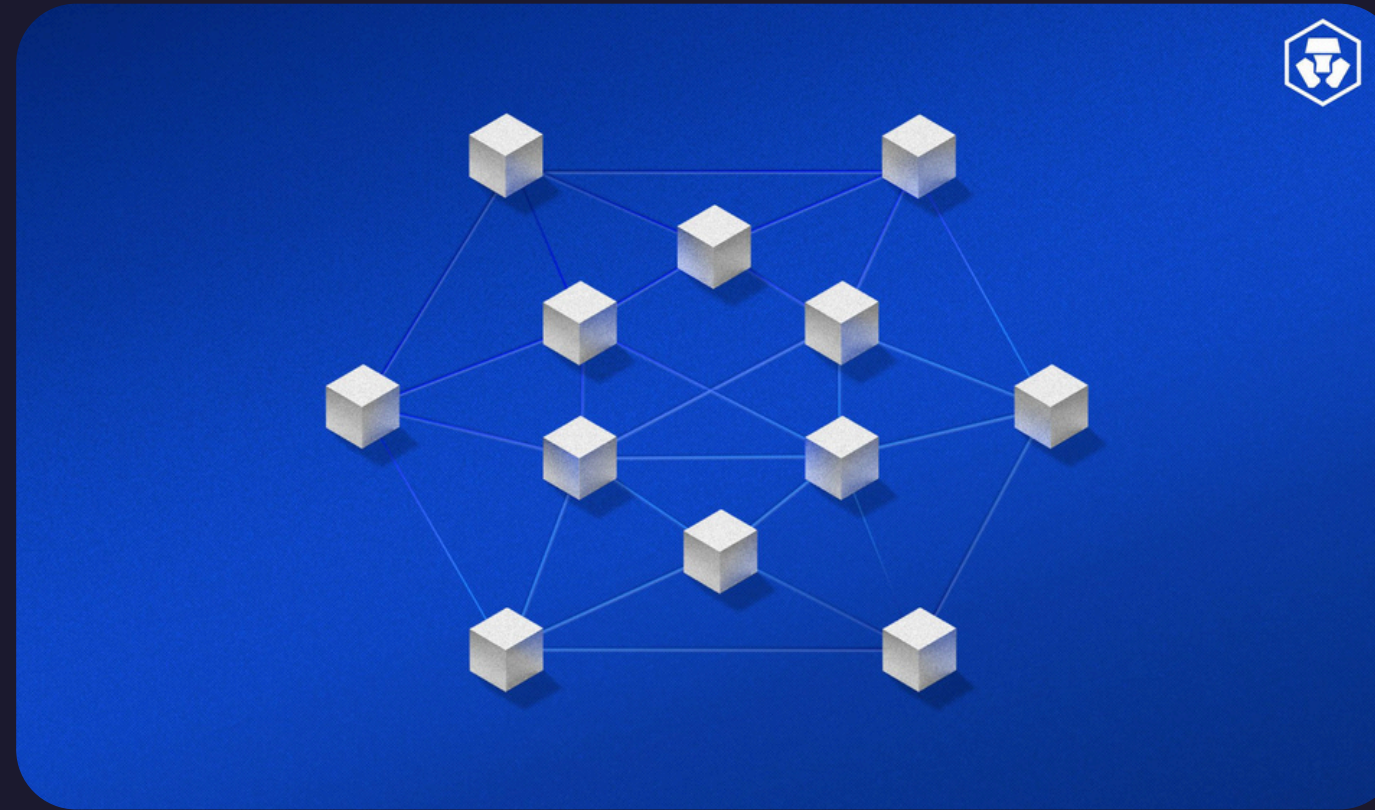


La soluzione



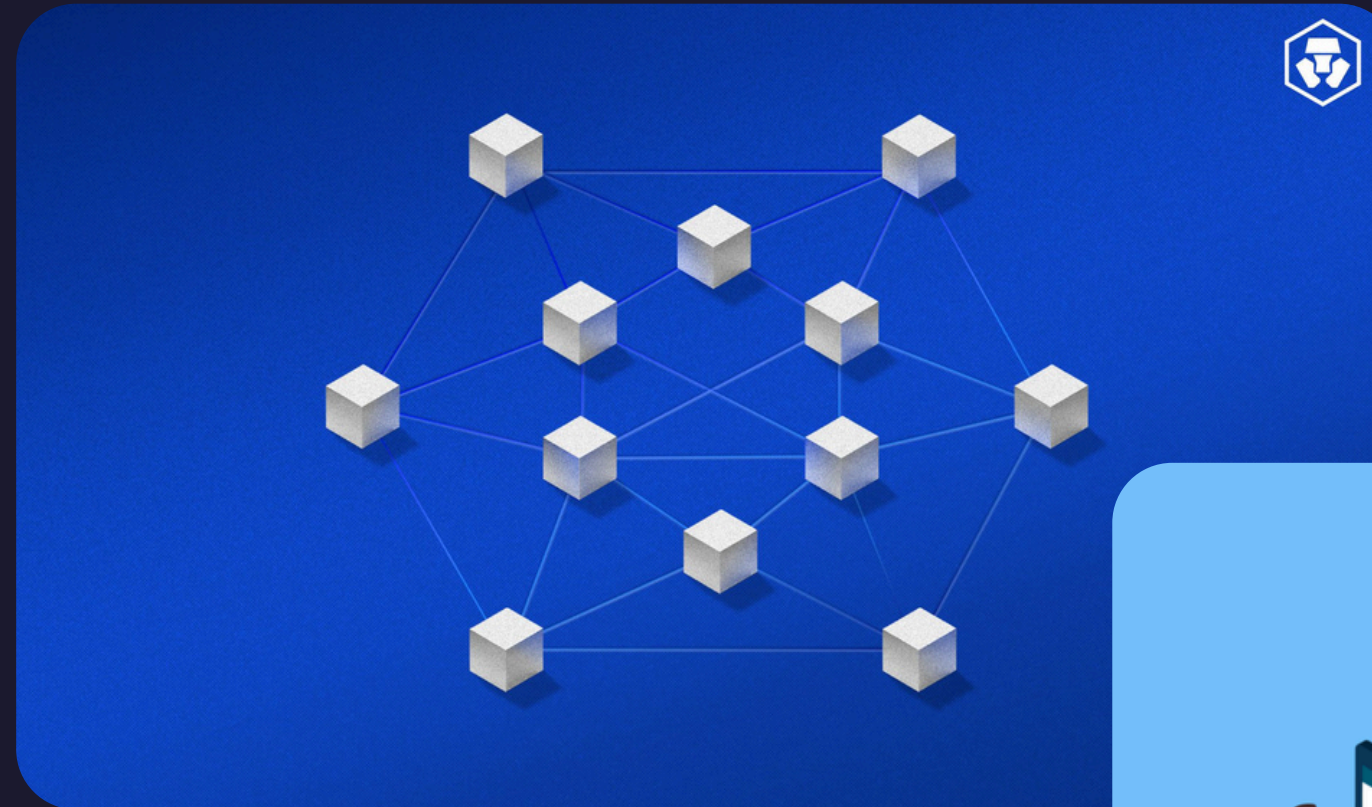
La soluzione

- Blockchain



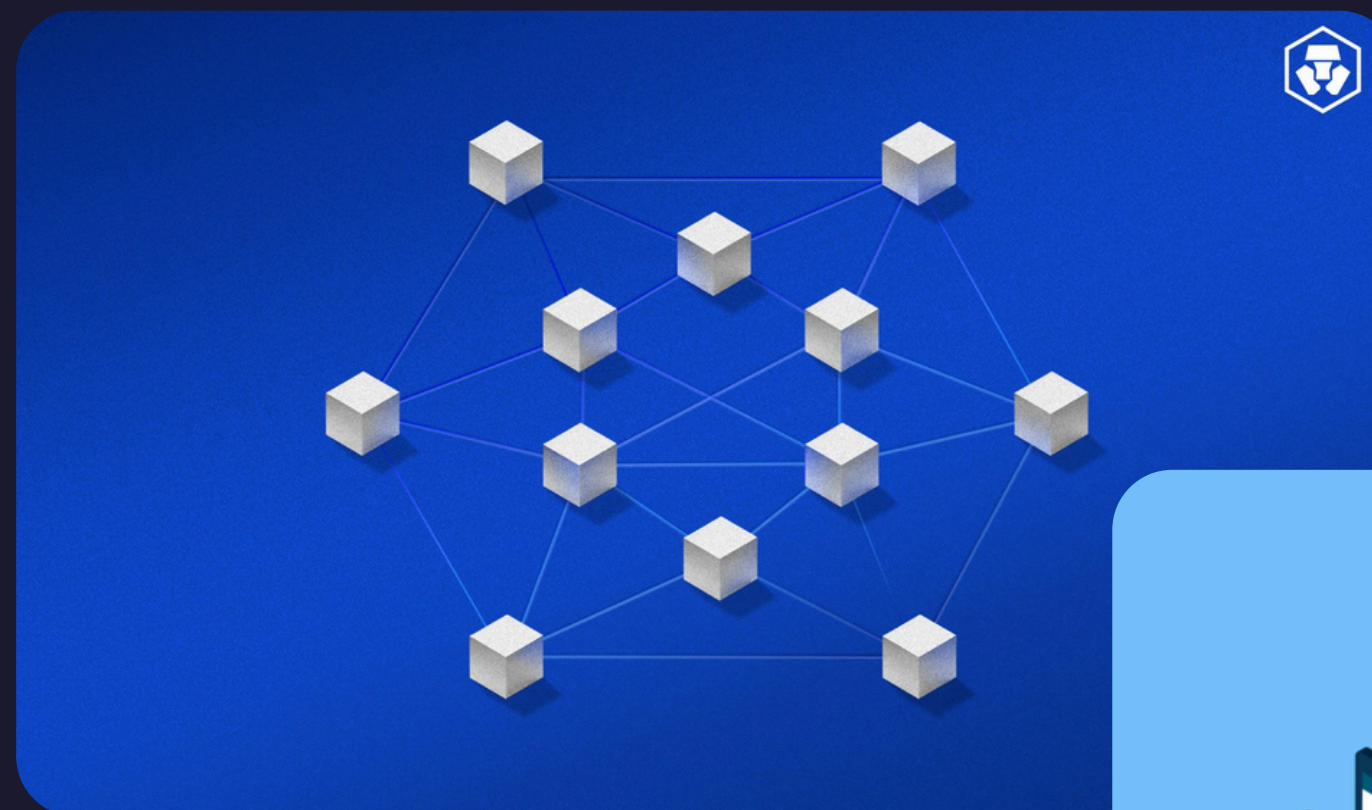
La soluzione

- Blockchain
- Smart contracts



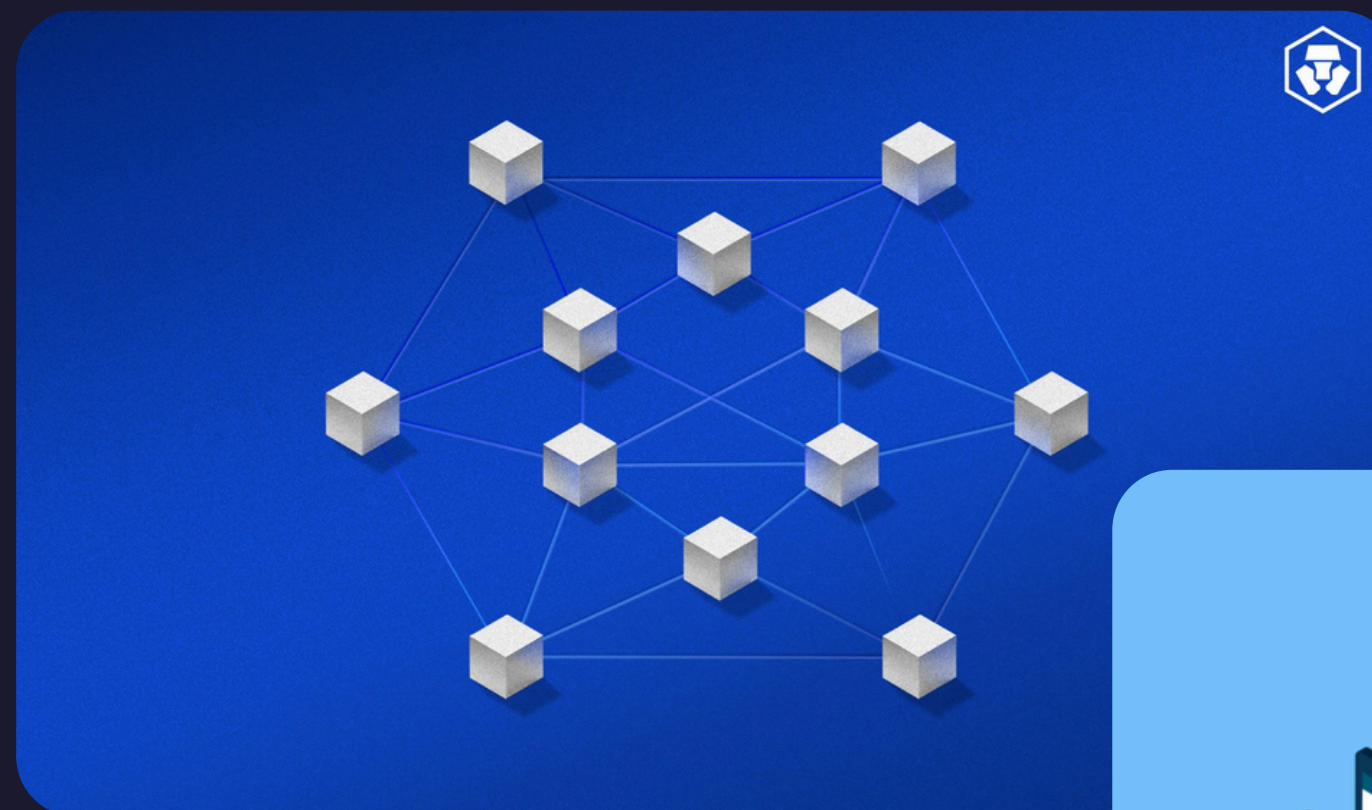
La soluzione

- Blockchain
- Smart contracts
- Unico garante



La soluzione

- Blockchain
- Smart contracts
- Unico garante
- Prove crittografiche





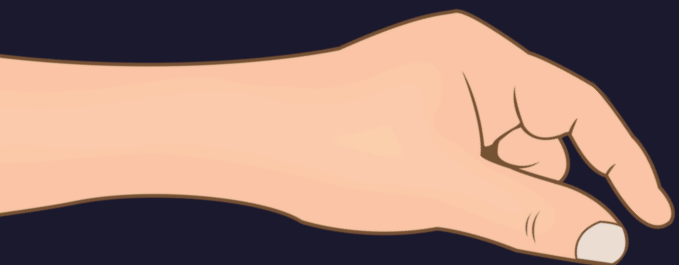
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Specifiche tecniche

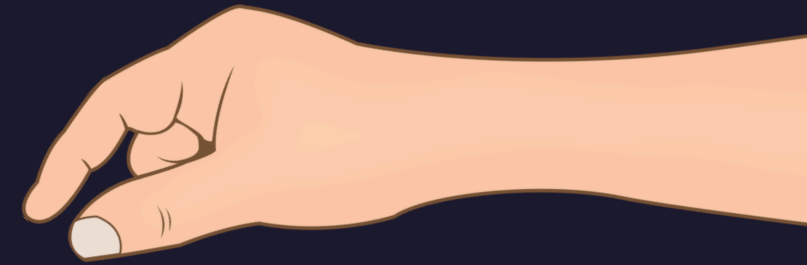
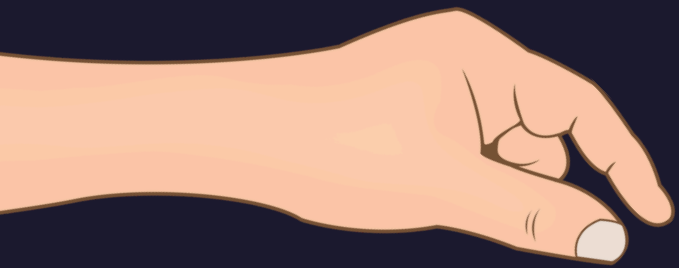
Disclaimer



Disclaimer



Disclaimer



Architettura

- Smart contracts
- ZKP
- Web Server



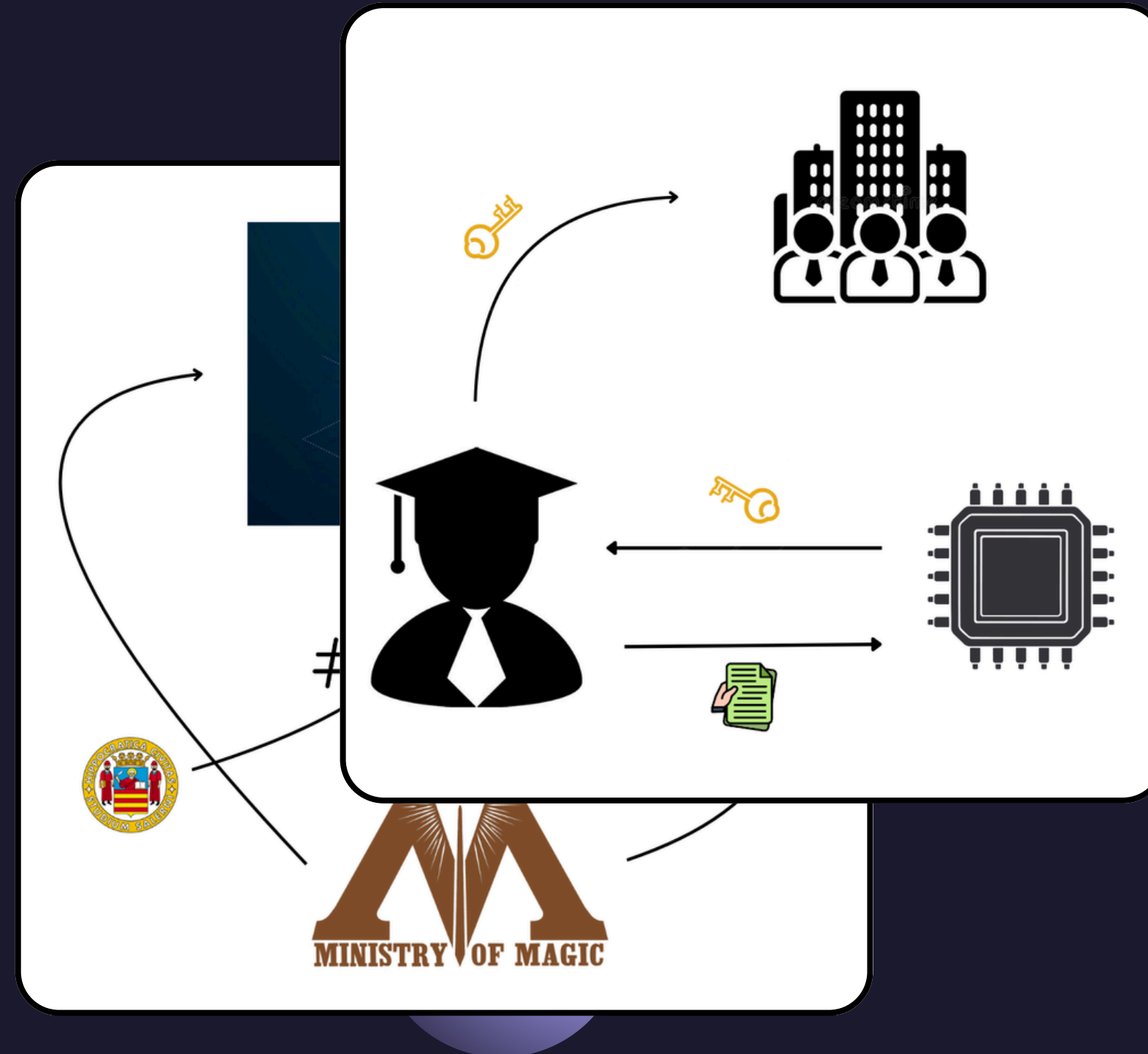
Architettura

- Smart contracts
- ZKP
- Web Server



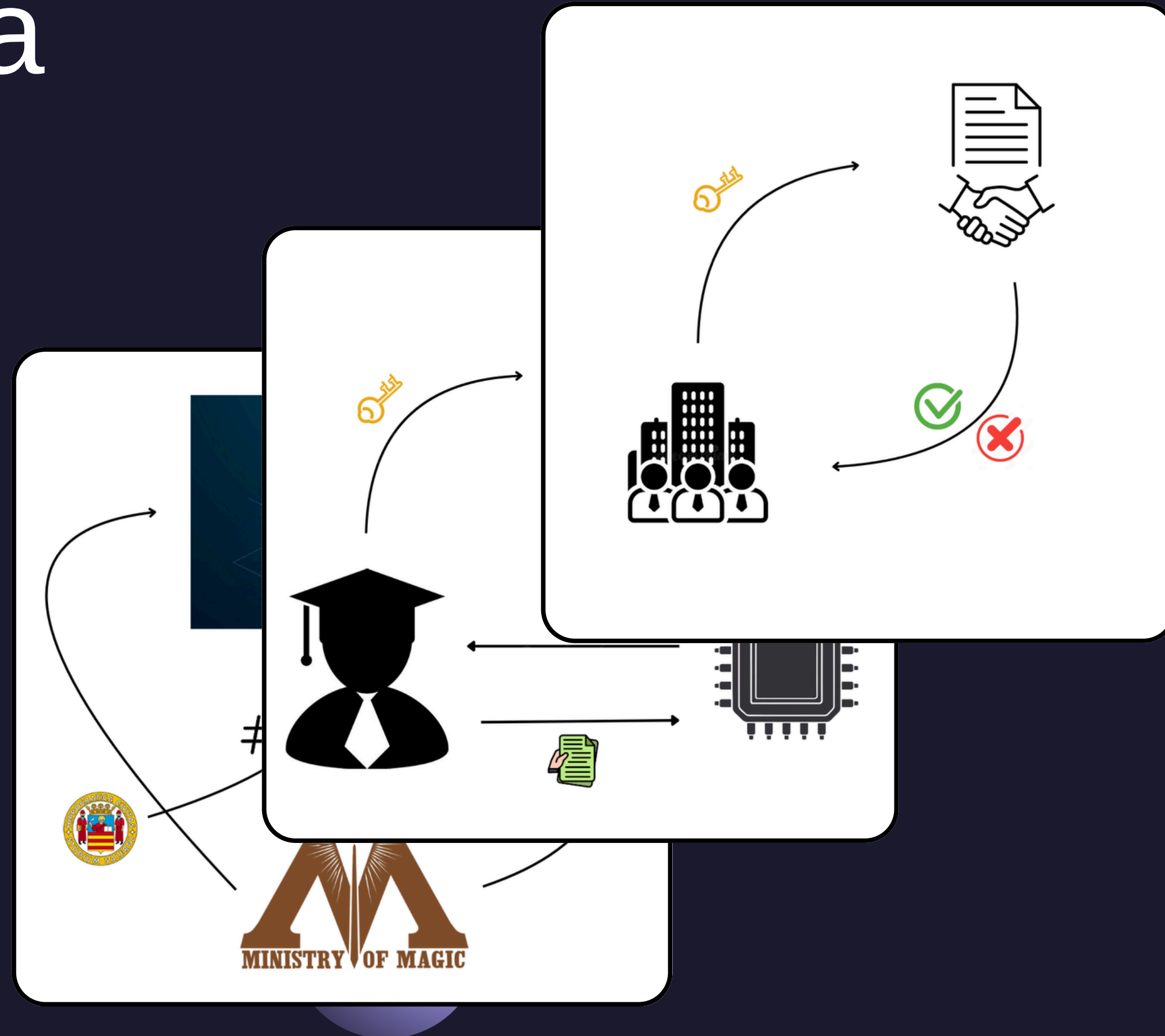
Architettura

- Smart contracts
- ZKP
- Web Server



Architettura

- Smart contracts
- ZKP
- Web Server



Blockchain

- permissioned
- pubblica



ÖREBRO
UNIVERSITET



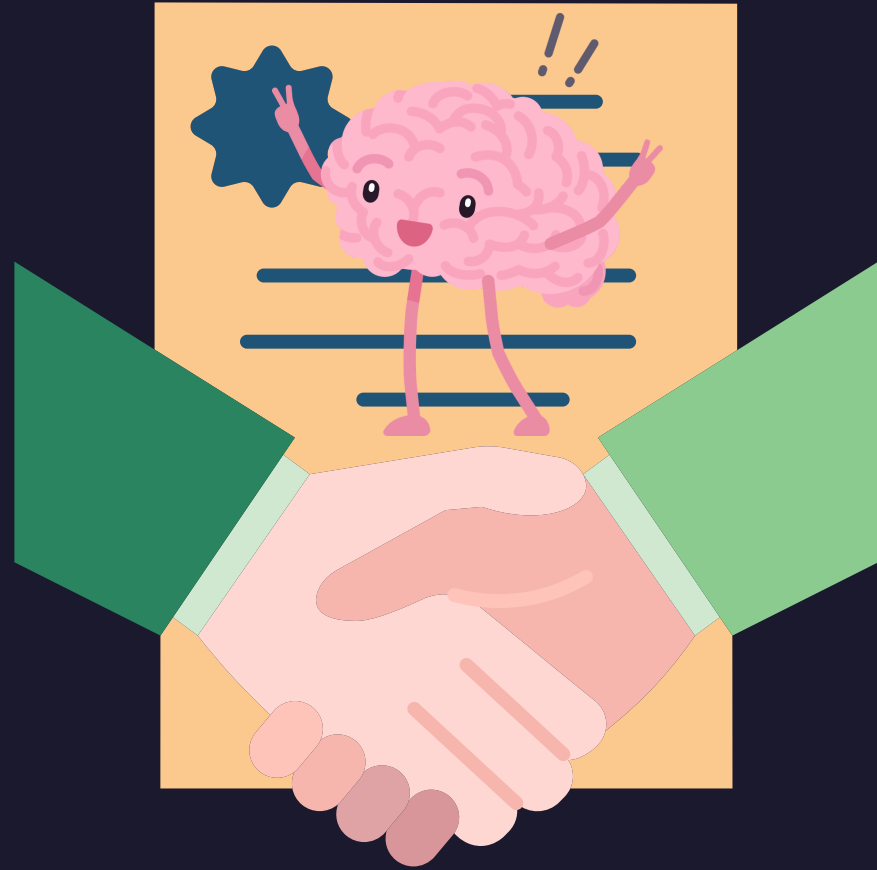
UNIVERSITY *of*
NICOSIA



NTNU



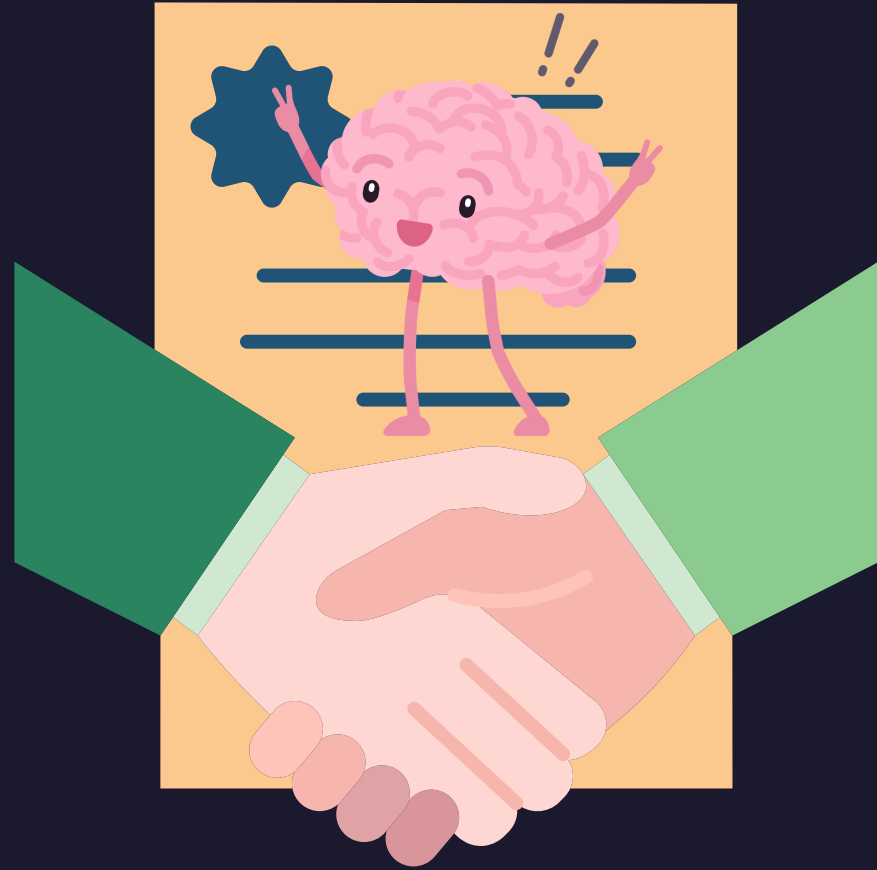
Smart Contract



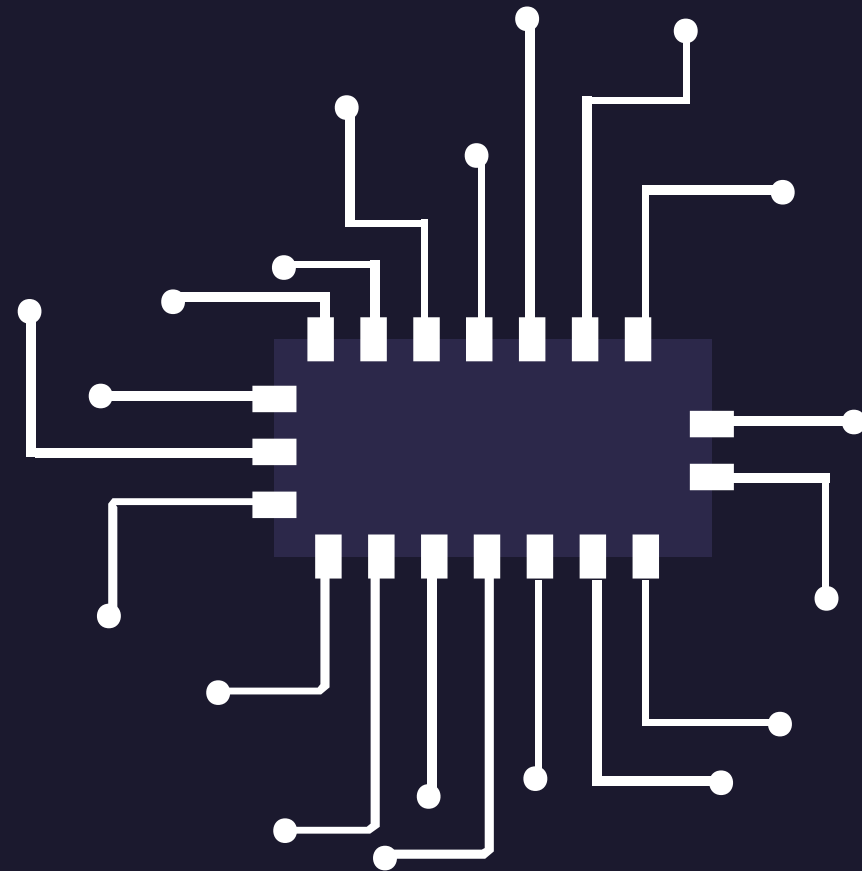
Authority.sol



Smart Contract



CircuitRegistry.sol



Smart Contract



CredentialIssuer.sol



Cosa si può provare con ZKP?

- voto di laurea superiore ad una certa soglia
- media esami superiore ad una certa soglia
- possesso titolo di studio in un certo ambito
- livello di istruzione



European
Qualifications
Framework

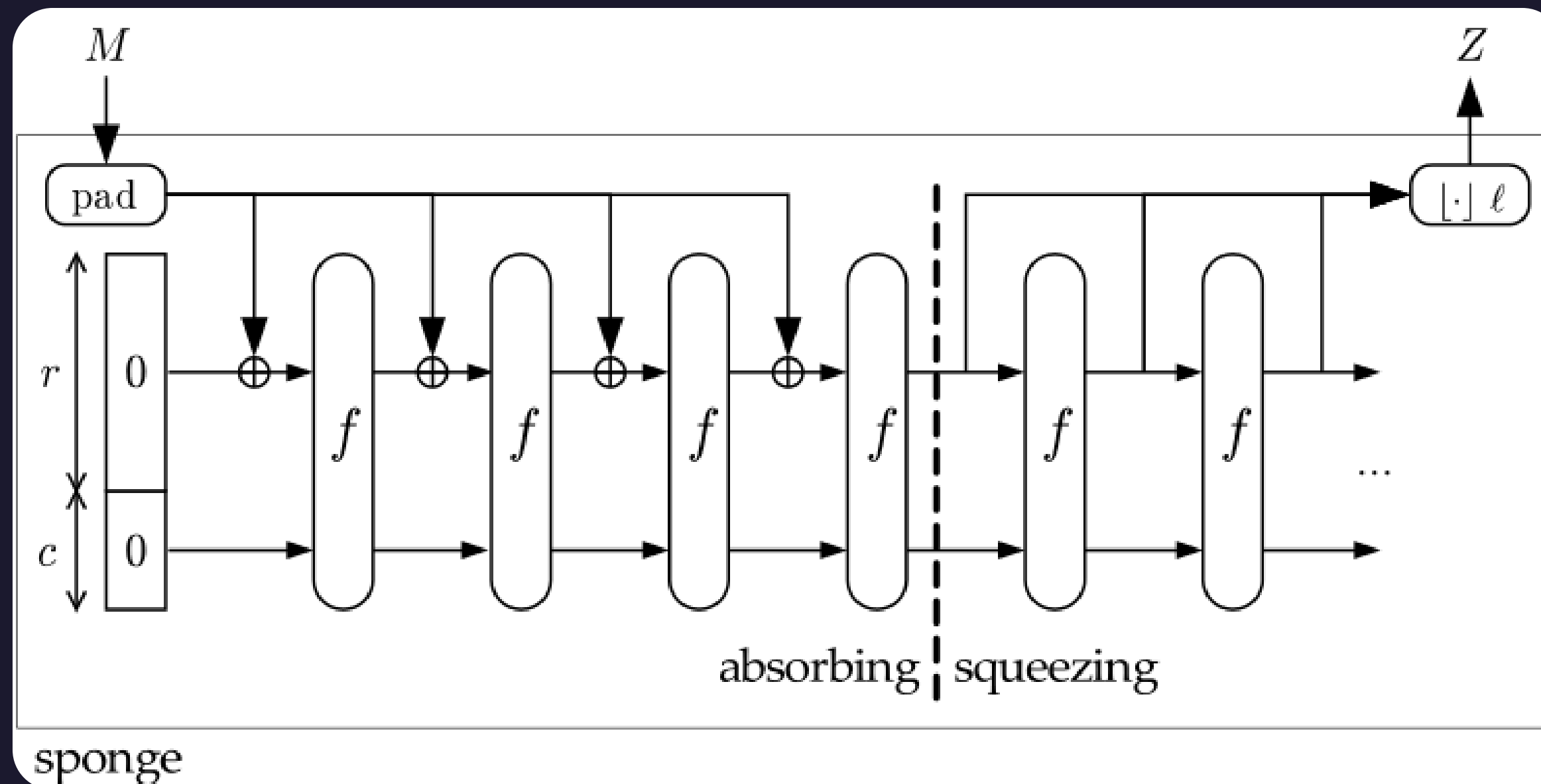
Formato Laurea

- informazioni circa il titolo di studio



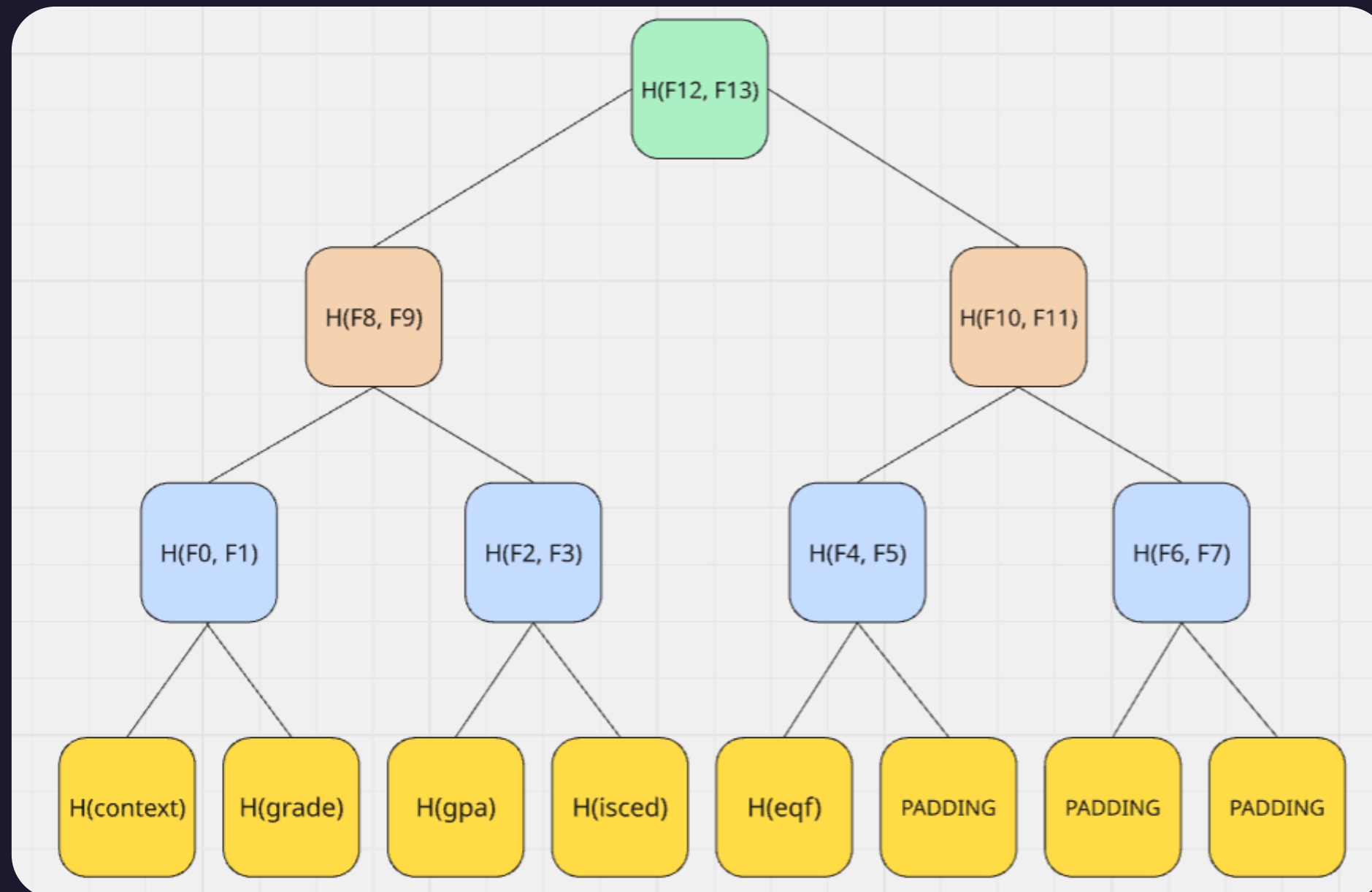
Formato Laurea

- informazioni circa il titolo di studio
- salt (keccak a 256 bit)

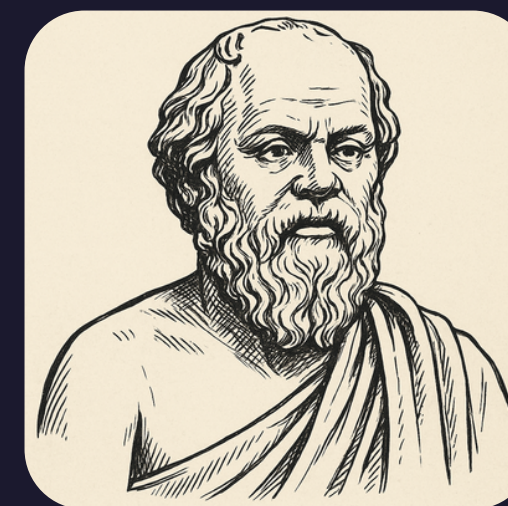
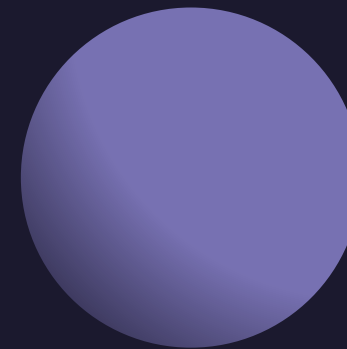


File laurea.json

- informazioni circa il titolo di studio
- salt (keccak a 256 bit)
- Merkle Tree (keccak a 256 bit)



Tecnologie usate



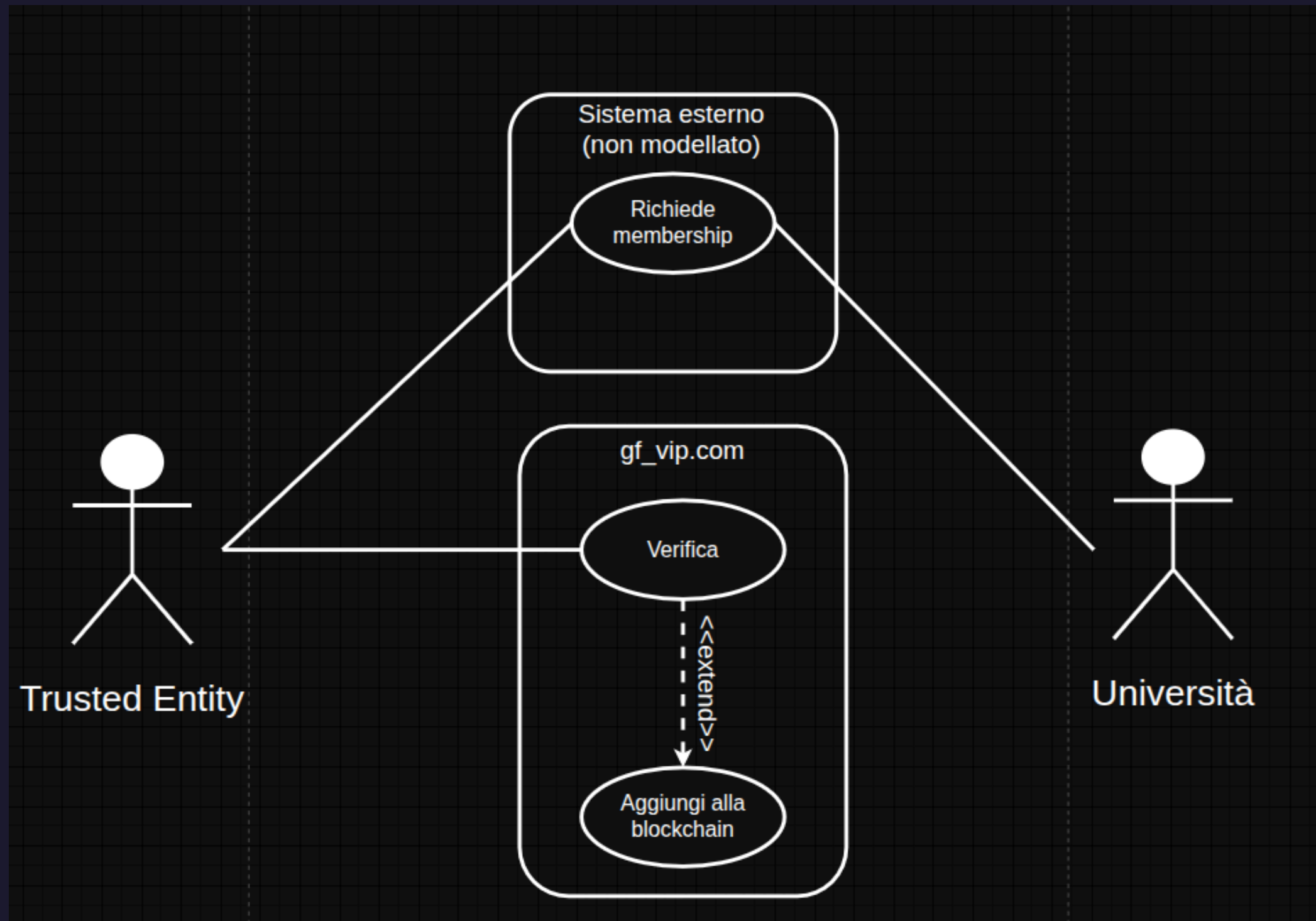
METAMASK™



Aspetti di ingegneria del software

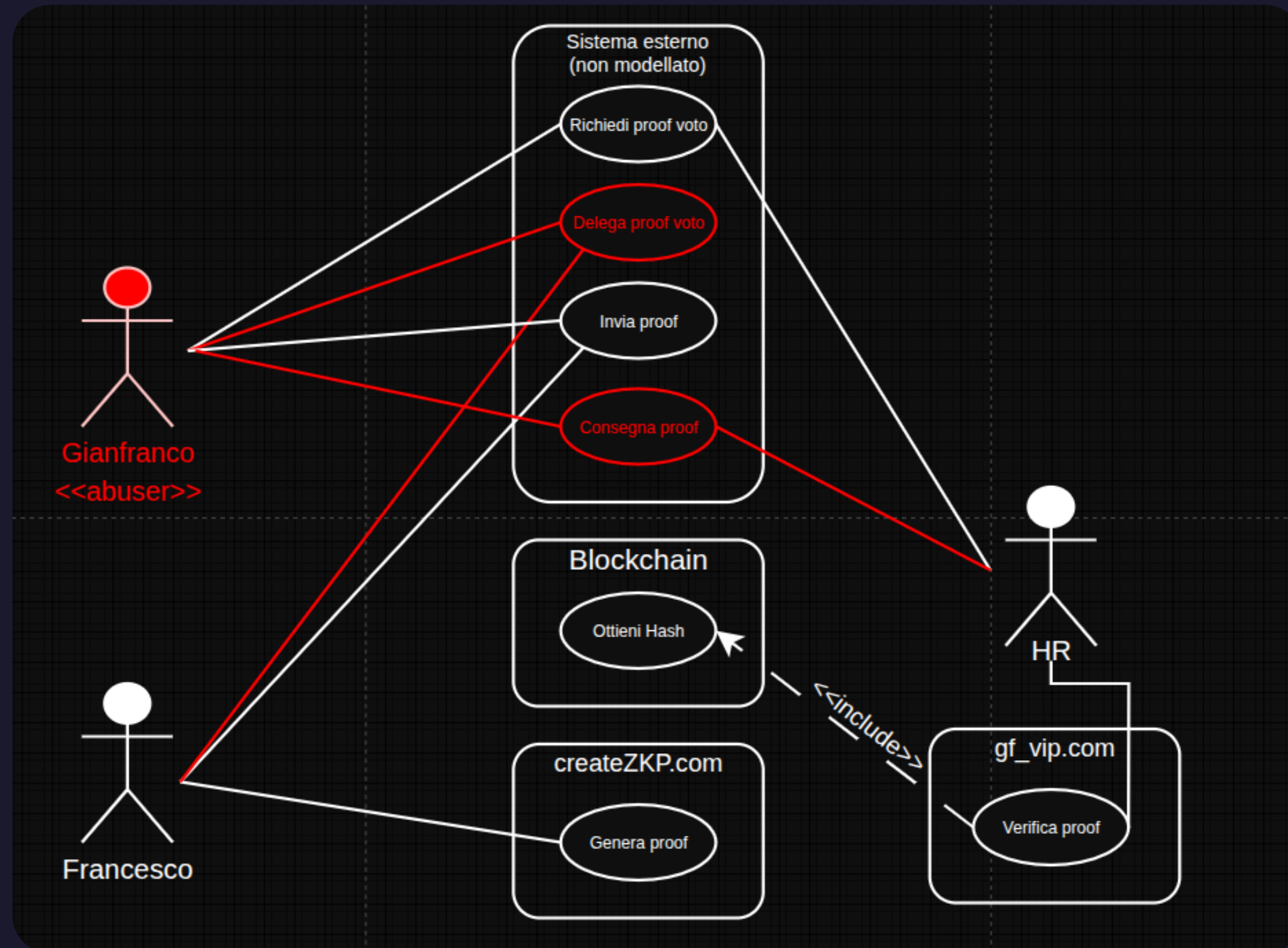
Use case

Inserimento Università



Abuse case

Proxy Attack



Abuse case

Mitigazioni

OGGI



Abuse case

Mitigazioni

OGGI



IN FUTURO





Grazie per l'attenzione!